

SANTHOSH KUMAR NEELI

GenAI Developer | AI Automation Engineer | LLM Applications | FastAPI | Playwright
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PROFESSIONAL SUMMARY

GenAI Developer with 3+ years of professional experience building AI-powered applications that reduce manual effort and improve delivery efficiency in QEA and enterprise automation. Hands-on experience designing LLM-driven solutions including test script generation, test data generation, XPath auto-healing, text-to-SQL, and chatbot-based automation workflows. Strong foundation in Python, FastAPI, TypeScript, React, Playwright, LangChain, LangGraph, MCP, vector databases, and multi-provider LLM integrations including Azure OpenAI, Copilot Bridge, Claude, and DeepSeek. Known for end-to-end ownership across architecture, frontend and backend development, orchestration, prompt design, framework analysis, and production-oriented automation tooling.

CORE SKILLS

Languages Python, TypeScript, JavaScript, Java, SQL	GenAI & LLM Generative AI, Prompt Engineering, RAG, Agentic AI, LangChain, LangGraph, OpenAI, Azure OpenAI, MCP	Frameworks FastAPI, Playwright, Selenium, POM, REST APIs, DOM Analysis, XPath Validation
Data & Infra Vector Databases, MySQL, Git, Excel/CSV Generation, API Integration	LLM Integrations Azure OpenAI, Copilot Bridge, Claude, DeepSeek, role-based prompting, provider abstraction	Domains AI Automation, Test Script Generation, Test Data Generation, Text-to-SQL, Oracle enterprise workflows

PROFESSIONAL EXPERIENCE

GenAI Developer / Software Development Engineer in Test (SDET) | Cognizant Technology Solutions | Bengaluru, India
June 2023 - Present

- Independently designed and developed GenAI applications for QEA workflows with a focus on reducing manual effort and improving automation delivery across Oracle enterprise use cases.
- Built an AI-powered test script generator that converts recorded user interactions into framework-aware, Page Object Model based automation assets for Playwright, Selenium, and Python workflows.
- Created a recorder-driven automation pipeline that captures structured user actions, business labels, DOM snapshots, and page metadata to support locator generation, validation, self-healing, and script generation.
- Engineered an XPath auto-healing workflow combining rule-based validation, DOM analysis, offline verification, and LLM-based fallback generation to repair unstable locators and automate locator updates.
- Developed a template-based test data generation platform with configurable business rules, dynamic columns, React-based user flows, and Excel/CSV output for complex enterprise test scenarios.
- Implemented a modular LLM integration layer with role-based prompts and provider abstraction, enabling seamless switching across Azure OpenAI, Copilot Bridge, Claude, and DeepSeek models.
- Built chatbot-style automation assistants, React-based user interfaces, and text-to-SQL style workflows to help users trigger automation tasks and retrieve structured outputs through natural language.
- Contributed across architecture, React frontend design, backend services, orchestration logic, prompt design, framework analysis, validation pipelines, and production-ready code structure.

- Worked on Selenium-based automation using Java and MySQL, contributing to functional test automation and reusable test design.
- Strengthened foundations in automation testing, debugging, and enterprise QA delivery practices before transitioning into AI-powered automation development.

SELECTED PROJECTS

AI-Powered Test Script Generator | Python, FastAPI, TypeScript, React, Playwright, Selenium, POM, Multi-LLM

- Developed a generator that transforms recorded browser flows into maintainable automation code and framework-aligned POM structures.
- Added framework analysis so generated scripts align with target project conventions, locator files, page files, and reusable methods.
- Designed a React-based frontend to guide users through flow upload, framework selection, and generated asset review.

Custom Recorder + DOM Evidence Layer | Playwright, TypeScript, React, JSON, DOM Snapshots

- Built a custom recorder inspired by Playwright codegen to capture clicks, fills, selects, changes, keydowns, business labels, outer HTML, page URLs, and DOM snapshots for each step.
- Designed a React-based frontend and review experience for inspecting captured steps, metadata, and evidence used in downstream automation generation.

XPath Auto-Healing Engine | Python, TypeScript, React, DOM Analysis, LLM Integration

- Created a locator repair workflow that validates generated selectors against stored DOM snapshots and falls back to LLM-assisted recovery when rule-based approaches fail.
- Designed a React-based interface for failure review, healing suggestions, and validation feedback to reduce maintenance effort for dynamic UI changes.

GenAI Test Data Generator & Automation Chatbot | Python, React, Excel/CSV, Vector DB, MCP, Azure OpenAI, Copilot Bridge

- Built a test data generation platform using templates, business rules, dynamic column handling, and export to Excel/CSV for high-volume test datasets.
- Integrated Azure OpenAI, Copilot Bridge, Claude, and DeepSeek support so users can work with the LLM provider available in their environment.
- Designed a React-based frontend and chatbot-driven flows, including text-to-SQL style capabilities, to let users request automation tasks and structured outputs conversationally.

EDUCATION

Bachelor of Technology in Electrical and Electronics Engineering | AITS, Rajampet | 2019 - 2023

Diploma in Electrical and Electronics Engineering | Govt Polytechnic, Kalikiri | 2016 - 2019

SSC | Zilla Parishad High School | 2015 - 2016

CERTIFICATIONS

- Programming in Java - IIT Kharagpur (NPTEL)
- Problem Solving Through Programming in C - IIT Kharagpur (NPTEL)